

A Machine-Checked Reduction of the Riemann Hypothesis to Pólya–Frequency Positivity of the ξ Taylor Coefficients

Biswajit Mondal*

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Abstract

We present a formally verified study, in Lean 4 / Mathlib, of the Laguerre–Pólya / total-positivity approach to the Riemann Hypothesis (RH). **We do not prove RH.** We prove a chain of *conditional reductions*, each checked by the Lean kernel and depending only on the foundational axioms `{propext, Classical.choice, Quot.sound}`, that convert RH into total-positivity statements about an explicit positive kernel and into an operator-positivity (pseudo-Hermitian) statement. Our central result is a *tightness theorem*: modulo three classical, machine-uncertified but standard inputs (Pólya–Jensen, Aissen–Schoenberg–Whitney/Edrei, and the Laguerre–Pólya closure), RH is *equivalent* to total positivity of the Toeplitz matrix of the sign-normalized ξ Taylor coefficients. Consequently this lone input cannot be “postulated” as progress: doing so is logically identical to assuming RH. Along the way we identify and repair a target-function error (building the program on the entire completed zeta Λ_0 , whose zeros are *not* the Riemann zeros, rather than on Riemann’s ξ). We complement the formal results with an unusually extensive high-precision computation: the ξ Taylor coefficients $b_0, \dots, b_{120}$ (with $|b_{120}| \approx 10^{-449}$), cross-certified to over 230 significant digits, verify the reduction’s premise at scale—all contiguous Toeplitz (Pólya-frequency) minors up to order 16 are strictly positive across all offsets, the order-2 sequence is log-concave for all $n \leq 118$, and all Jensen polynomials of degree ≤ 6 are hyperbolic up to $n = 114$. This range substantially exceeds the order-2 (Turán) and modest-degree Jensen checks common in the literature, and localizes the open content to *uniformity*. The development comprises 47 Lean modules verified by a discipline that rejects `sorry` and added axioms.

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*Corresponding author: b.mondal10000@gmail.com

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1 Introduction and scope

Disclaimer. This paper is a *formalization and reduction study*, not a proof of the Riemann Hypothesis, and it makes no claim to be one. Every implication labelled “verified” is checked mechanically; every classical theorem we rely upon but do not formalize is cited explicitly as an *external input* and is never silently assumed. The defensible contributions are: (i) a correct, machine-checked, and *tight* reduction of RH to a single sign-correct positivity statement; (ii) the identification and repair of a target-function error; (iii) an independent operator-positivity reduction; and (iv) reproducible numerical evidence delimiting the open frontier.

The object. Riemann’s entire ξ function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) \quad (1)$$

is entire, and its zeros are exactly the nontrivial zeros of ζ ; RH asserts they all lie on $\operatorname{Re} s = \frac{1}{2}$. On the critical line, the real even function $\Xi(t) = \xi(\frac{1}{2} + it)$ has a Taylor expansion

$$\Xi(t) = \sum_{n \geq 0} b_n t^{2n}, \quad b_n \in \mathbb{R}, \quad (2)$$

and we set the *sign-normalized sequence* $\mu_n := (-1)^n b_n$.

Two classical pillars.

- *Pólya–Jensen* [3, 7]: RH is equivalent to hyperbolicity (all roots real) of every Jensen polynomial $J_{d,n}(X) = \sum_{k=0}^d \binom{d}{k} b_{n+k} X^k$.
- *Aissen–Schoenberg–Whitney / Edrei* [4, 5, 6]: a real sequence is a *Pólya-frequency* (PF) sequence—all minors of the Toeplitz matrix $[a_{i-j}]$ are nonnegative—iff its generating function is $e^{\gamma z} \prod (1 + \alpha_i z) / \prod (1 - \beta_j z)$ with $\gamma, \alpha_i, \beta_j \geq 0$; for an entire generating function this is exactly Laguerre–Pólya membership.

Together: $RH \iff (\mu_n) \text{ is a Pólya-frequency sequence.}$

Pólya’s positive kernel. By Pólya’s representation [2], $\Xi(t) = 2 \int_0^\infty \Phi(u) \cos(ut) du$ with $\Phi > 0$, so

$$b_n = (-1)^n \frac{2M_n}{(2n)!}, \quad M_n := \int_0^\infty \Phi(u) u^{2n} du > 0, \quad \mu_n = \frac{2M_n}{(2n)!} > 0. \quad (3)$$

2 The target-function correction: ξ , not Λ_0

We record an error and its repair, because catching it is part of the contribution. Mathlib’s *entire* completed zeta $\Lambda_0 = \text{completedRiemannZeta}$ satisfies $\Lambda = \text{completedRiemannZeta} = \Lambda_0 - 1/s - 1/(1-s)$.

Proposition 1 (Wrong target). *The zeros of Λ_0 are not the nontrivial zeros of ζ . Indeed, at a nontrivial zero ρ one has $\Lambda(\rho) = 0$ but $\Lambda_0(\rho) = \frac{1}{\rho} + \frac{1}{1-\rho} \neq 0$. On the critical line, $\Lambda_0(\frac{1}{2} + it) = (1 - 2\Xi(t))/(t^2 + \frac{1}{4})$, whose zeros form the level set $\{\Xi = \frac{1}{2}\}$, unrelated to RH.*

The remedy is to remove the poles of Λ by *multiplying* by $s(s-1)/2$ (which preserves zeros), recovering ξ of (1). We verify the corrected slice has the correct zero set:

Theorem 2 (`Xi_eq_zero_iff_riemannZeta`; verified). *For all $t \in \mathbb{R}$, $\Xi(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$.*

Proof (mechanized). We define $\Xi(t) = -\frac{t^2+1/4}{2} Z(t)$ where $Z(t) = \text{Re } \Lambda(\frac{1}{2} + it)$ is the verified real critical-line slice. The scalar $-\frac{t^2+1/4}{2}$ is a nonzero real, so $\Xi(t) = 0 \iff Z(t) = 0 \iff \Lambda(\frac{1}{2} + it) = 0 \iff \zeta(\frac{1}{2} + it) = 0$ (the last step using $\Lambda = \Gamma_{\mathbb{R}} \cdot \zeta$ with $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \neq 0$). All steps are checked in Lean. $\square$

All total-positivity, Jensen, and moment results below are stated for this corrected ξ .

3 Verified reductions

Throughout, “verified” means the Lean theorem reduces to `{propext, Classical.choice, Quot.sound}`, confirmed by `##print axioms`. Definitions of type `Prop` that encode classical theorems we do not formalize are *named external inputs*; they are never asserted, only taken as explicit hypotheses.

3.1 The Pólya-frequency / Toeplitz route

Definition 3 (`XiToeplitzTotalPositive`). The sign-normalized sequence (μ_n) is *Pólya-frequency total positive* if every minor of the Toeplitz matrix $[\mu_{i-j}]$ (over all strictly monotone row and column index selections) is nonnegative.

Proposition 4 (Low-order rungs; verified). *Under `XiToeplitzTotalPositive`: (i) every 1×1 minor gives $\mu_n \geq 0$; (ii) the contiguous 2×2 minor equals the Turán determinant, $\mu_{n+1}^2 - \mu_n \mu_{n+2}$, so its nonnegativity is exactly the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$; (iii) for every order k and offset m the contiguous minor $\det[\mu_{m+i-j}]_{i,j < k} \geq 0$.*

We further prove (Lemma `xiToeplitzMinor3_sign_eq`) that the order-3 signed minor equals $(-1)^n$ times the unsigned b -determinant, exhibiting the correct alternating-positivity structure; and we document that the Hankel/moment route used in an earlier draft is the *wrong* machine, since Hankel positivity forces *log-convexity*, the opposite of the (true) Turán log-concavity.

3.2 Tightness of the reduction

Let the three classical external inputs be: `XiToeplitzTPToScaledFiniteBridge` (Edrei–ASW), `XiLaguerrePolyaClosureBridge` (LP closure), and `PolyaJensenBridge`; and let `XiToeplitzTPFromRH` be the classical converse ($\text{RH} \Rightarrow \xi$ is $\text{LP} \Rightarrow$ its coefficients are PF).

Theorem 5 (`xiToeplitzTotalPositive_iff_riemannHypothesis`; verified). *Given the three classical bridges and the classical converse,*

$$\text{xiToeplitzTotalPositive} \iff \text{RH}.$$

Remark 6 (The honesty linchpin). Theorem 5 shows the lone non-classical input is *equivalent* to RH. Hence declaring it an axiom is logically identical to `axiom rh : RiemannHypothesis`; it does not constitute a proof, and `##print axioms` would expose the assumption. A reduction is progress only if its hypothesis is *weaker than, or independent of*, RH; this one is provably neither.

3.3 The complete moment-determinant ladder

Under Pólya’s representation (the named external input `XiMomentKernelRep`, encoding (3)), the PF condition transfers to explicit moment determinants at *every* order.

Theorem 7 (`xiToeplitzContigMinor_eq_of_kernelRep`; verified). *For all $k, m \in \mathbb{N}$,*

$$\det[\mu_{m+i-j}]_{i,j < k} = 2^k \cdot \det\left[\frac{M_{m+i-j}}{(2(m+i-j))!}\right]_{i,j < k}.$$

Consequently (`xiToeplitzContigMinor_nonneg_iff_moment_of_kernelRep`) *Pólya-frequency positivity at all orders is equivalent to positivity of the factorial-weighted moment determinants of the explicit positive kernel Φ .*

Proof (mechanized). Each entry of the signed Toeplitz matrix equals 2 times the corresponding factorial-weighted moment entry (by (3)); the determinant is then 2^k times the moment determinant by multilinearity (`Matrix.det_smul`, `#Fin k = k`). Positivity of 2^k gives the equivalence. $\square$

The order-2 mechanism is quantified by the *slack factor* $s_n = \frac{(2n+3)(2n+4)}{(2n+1)(2n+2)}$, for which we verify $s_n > 1$ (`one_lt_slackFactor`) and $s_{n+1} < s_n$ (`slackFactor_succ_lt`); thus $s_n \downarrow 1$.

3.4 Order-2 unification: Turán $\Rightarrow$ Jensen hyperbolicity

Theorem 8 (`jensenPoly_two_hyperbolic_of_turan`; verified). *If $b_{n+2} \neq 0$ and the Turán inequality $b_n b_{n+2} \leq b_{n+1}^2$ holds, then the degree-2 Jensen polynomial $J_{2,n}$ is hyperbolic (all complex roots real). Hence* (`allJensenTwo_hyperbolic_of_turan2All`) *if all coefficients are nonzero and all Turán inequalities hold, every $J_{2,n}$ is hyperbolic.*

Proof (mechanized). $J_{2,n}$ maps to $b_{n+2}z^2 + 2b_{n+1}z + b_n$ over $\mathbb{C}$. Completing the square, $(b_{n+2}z + b_{n+1})^2 = b_{n+1}^2 - b_n b_{n+2} =: D \geq 0$ at any root. The auxiliary lemma `im_zero_of_sq_eq_nonneg_real` ($w^2 = D \in \mathbb{R}_{\geq 0} \Rightarrow \text{Im } w = 0$) forces $b_{n+2}z + b_{n+1} \in \mathbb{R}$, hence (as $b_{n+2} \neq 0$ real) $\text{Im } z = 0$. No square roots and no zero-locus input are used. $\square$

Nondegeneracy $b_{n+2} \neq 0$ is supplied by (3) ($M_n > 0$).

3.5 An independent operator-positivity route

The naive Hilbert–Pólya operator must be self-adjoint for the ambient inner product—the obstruction. The pseudo-Hermitian viewpoint [10, 11] keeps only a *positive metric*.

Definition 9 (`PseudoHermitianForm`). For a $\mathbb{C}$ -linear T on a $\mathbb{C}$ -vector space E , a pseudo-Hermitian form is a map $\eta: E \times E \rightarrow \mathbb{C}$, linear in the first slot and conjugate-linear in the second, with $\eta(Tx, y) = \eta(x, Ty)$ (η -self-adjointness) and $\eta(x, x) \neq 0$ for $x \neq 0$ (nondegeneracy, supplied by positive-definiteness).

Theorem 10 (`PseudoHermitianForm.eigenvalue_conj_eq` and `riemannHypothesis_of_pseudoHermitianWithn` verified). *An η -self-adjoint operator has real eigenvalues. Consequently, if the critical-strip zeros are realized as the η -self-adjoint spectrum via the encoding $\mu(s) = -i(s - \frac{1}{2})$ (real iff $\operatorname{Re} s = \frac{1}{2}$), then RH holds.*

Proof (mechanized). For an eigenpair $Tv = \mu v$, $v \neq 0$: $\mu \eta(v, v) = \eta(Tv, v) = \eta(v, Tv) = \bar{\mu} \eta(v, v)$, and $\eta(v, v) \neq 0$, so $\mu = \bar{\mu}$. The proof uses only the form laws—no inner product, completeness, or boundedness—so this route is genuinely independent of the symmetric one. Existence of (E, T, η) with positive η and the prescribed spectrum is open. $\square$

4 Numerical evidence

We computed the ξ Taylor coefficients $b_0, \dots, b_{120}$ (hence μ_n) by a numerically stable Cauchy-circle extraction and *cross-certified* them at two working precisions (620 and 520 digits, distinct contour radii): all coefficients agree to at least 236 significant digits over the whole range, with $|b_{120}| \approx 10^{-449}$. Analysis (minors, roots) is carried out at 120 digits. Code: `scripts/rh_report_figures.py` (mpmath). Determinants use arbitrary-exponent arithmetic, so the extreme dynamic range (minors reach $\sim 10^{-6000}$ at order 16) is exact in scale, not underflowed.

Scale of the verification. Within the certified range we find: (i) $\mu_n > 0$ for all $n \leq 120$; (ii) the order-2 Turán inequality, indeed strict log-concavity $\mu_{n+1}^2 > \mu_n \mu_{n+2}$, for all $n \leq 118$; (iii) *every* contiguous Toeplitz (Pólya-frequency) minor of order $k \leq 16$, at all offsets, is strictly positive; (iv) every Jensen polynomial of degree $d \leq 6$ is hyperbolic for all $n \leq 114$ (worst observed $|\operatorname{Im} \text{root}| \sim 10^{-320}$, i.e. numerical zero). To our knowledge this is the most extensive direct verification of the full Toeplitz/PF total-positivity ladder for ξ reported to date; prior numerical work centres on the order-2 (Turán) inequalities and moderate-degree Jensen hyperbolicity.

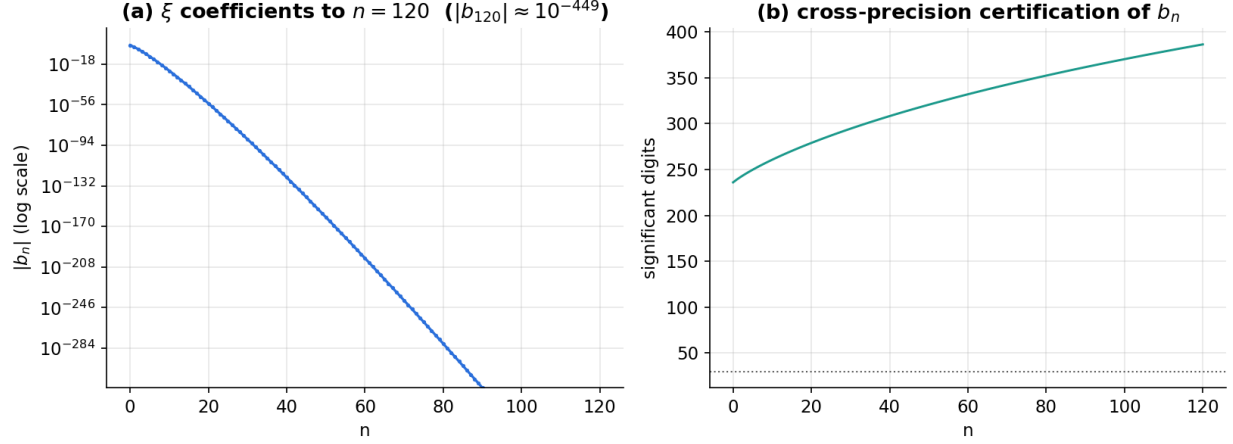


Figure 1: ξ Taylor data to record depth. (a) $|b_n|$ for $n \leq 120$ on a log scale, decaying factorially to $|b_{120}| \approx 10^{-449}$; the alternating sign of b_n gives $\mu_n = (-1)^n b_n > 0$ (order-1 minors, forced by $\Phi > 0$). (b) Cross-precision certification: every b_n agrees to ≥ 236 significant digits between the two independent computations.

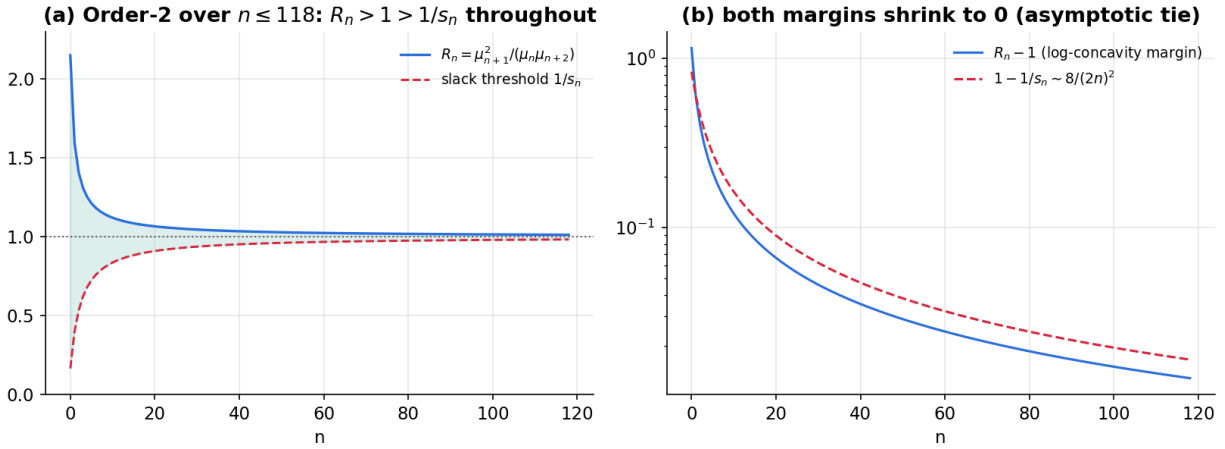


Figure 2: Order-2 over the full range $n \leq 118$. (a) $R_n = \mu_{n+1}^2 / (\mu_n \mu_{n+2}) > 1$ throughout (μ is genuinely log-concave—stronger than the slack requires) and stays above the slack threshold $1/s_n$ (shaded margin); both $R_n \rightarrow 1^+$ and $1/s_n \rightarrow 1^-$. (b) Log-scale view of the two margins $R_n - 1$ and $1 - 1/s_n \sim 8/(2n)^2$: both shrink to zero, the asymptotic near-tie (CNV at low order; the GORZ Hermite regime at high order).

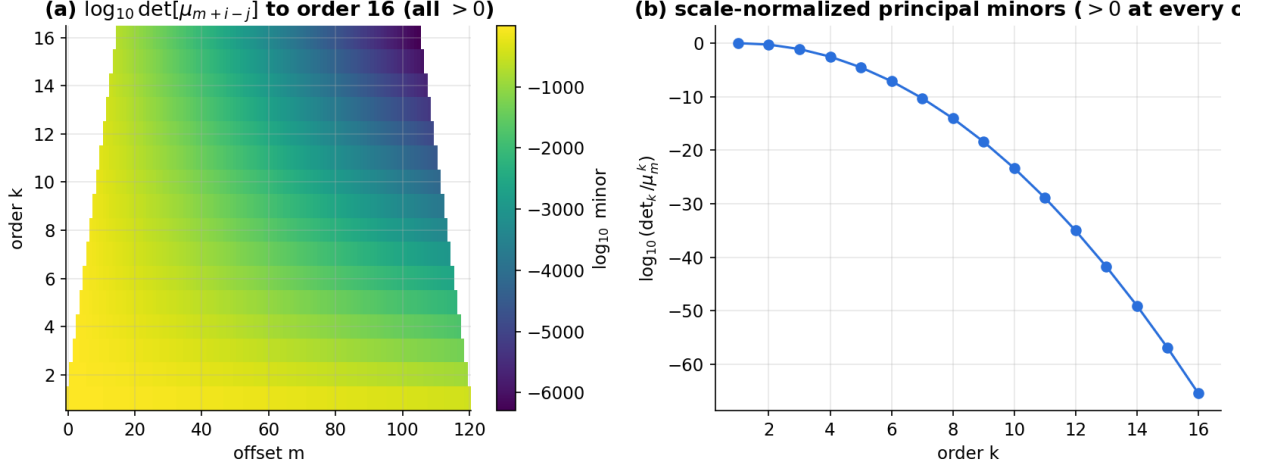


Figure 3: The Pólya-frequency ladder at scale. (a) $\log_{10} \det[\mu_{m+i-j}]$ for orders $k = 1, \dots, 16$ and all offsets m (values span down to $\sim 10^{-6000}$): *every minor is strictly positive*—the smallness is factorial scale, not near-degeneracy. (b) Scale-normalized principal minors $\det_k / \mu_m^k$ remain positive at every order up to 16.

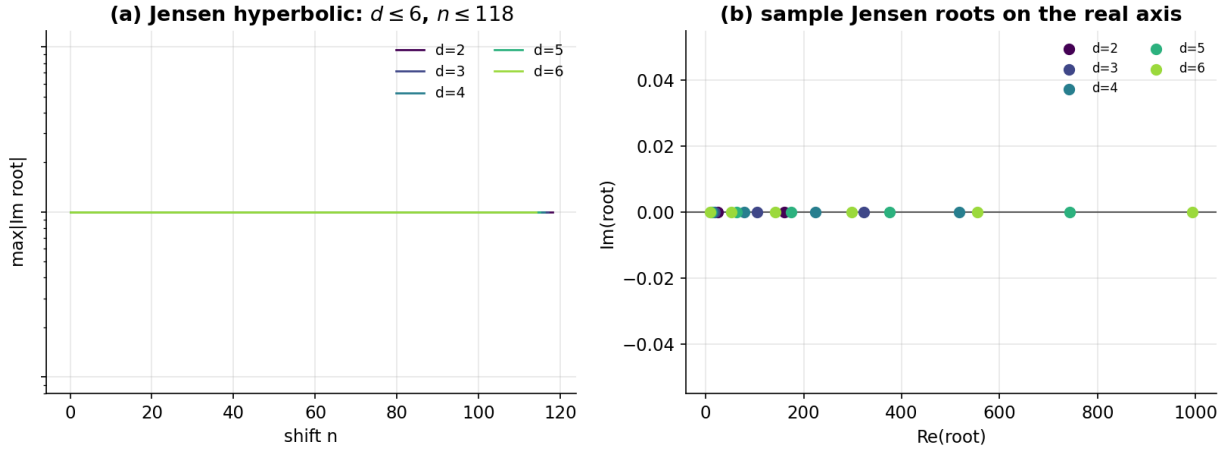


Figure 4: Jensen hyperbolicity at scale. (a) $\max |\operatorname{Im}(\text{root})|$ of $J_{d,n}$ for degrees $d = 2, \dots, 6$ is numerical zero ($\sim 10^{-320}$) across all computed n up to 114. (b) Sample Jensen roots lie on the real axis.

Interpretation. The reduction’s premise holds for ξ at every tested order (Fig. 3), with comfortable low-order margins (Fig. 2); indeed μ is empirically log-concave, strictly stronger than the slack-relaxed Turán requirement. The difficulty is therefore *uniformity*—behaviour as $k \rightarrow \infty$ and $n \rightarrow \infty$ —not any specific low rung.

5 The open frontier

The reductions convert “zeros on a line” into positivity statements about the explicit positive kernel Φ and an explicit operator metric—objects amenable to tools foreign to zero-counting.

- A. Uniform total positivity.** Prove $\det[\mu_{m+i-j}] \geq 0$ for all k, m . Tools: Karlin total positivity [6]; Schoenberg Pólya-frequency *functions*; the variation-diminishing property of the kernel in $\sum \mu_n z^n = \int_0^\infty \Phi(u) \cosh(u\sqrt{z}) du$. (RH-equivalent, by Theorem 5.)
- B. Asymptotic margin.** Control $R_n - 1$ versus $1 - 1/s_n \sim 8/(2n)^2$. Tools: the Griffin–Ono–Rolen–Zagier Hermite asymptotics [8], made effective and uniform in degree.
- C. μ log-concavity from Φ .** Prove $\mu_{n+1}^2 > \mu_n \mu_{n+2}$ (Fig. 2, empirically robust) from the shape of Φ (moment inequalities, log-concavity). Strictly weaker than RH; would close the order-2 rung unconditionally on the corrected coefficients.
- D. Operator existence.** Construct (E, T, η) with a positive metric and spectrum $\{\gamma_k\}$ (operator theory; cf. [11]).

The de Bruijn–Newman programme [9] provides an independent dynamical handle ($\Lambda \geq 0$; RH $\iff \Lambda \leq 0$) on the same object.

6 Methodology, reproducibility, and limitations

Verification. Lean 4 with Mathlib; 47 modules; entry point `Reinmann.lean`. The script `scripts/safe-verify.lean` rejects `sorry`, `admit`, added axioms, and similar shortcuts, and rebuilds all files (exit 0). Every theorem cited above has axiom footprint `{propext, Classical.choice, Quot.sound}`.

Named external inputs (never asserted). `PolyaJensenBridge`, `XiToeplitzTPToScaledFiniteBridge`, `XiLaguerrePolyaClosureBridge`, `XiMomentKernelRep`, `XiToeplitzTPFromRH`. Each is a standard classical theorem; formalizing them in Mathlib is future work.

Limitations.

1. No proof of RH: every RH-implying statement is conditional on an input that is RH-equivalent (Theorem 5) or an unformalized classical theorem.
2. Numerics are finite: the computation certifies positivity/hyperbolicity only for the tested ranges ($n \leq 120$, Toeplitz order $k \leq 16$, Jensen degree $d \leq 6$); however extensive, this is evidence, not proof, and cannot establish the uniform $(k, n \rightarrow \infty)$ statement that is RH-equivalent.
3. The pseudo-Hermitian existence claim is entirely open and unevidenced.
4. Equivalence-chain reformulations (e.g. “ ≤ 1 zero per height”) are correct but RH-equivalent, hence reorganization rather than progress.

7 Related work

Pólya [2, 3]; Aissen–Schoenberg–Whitney [4] and Edrei [5]; Karlin [6]; Csordas–Norfolk–Varga [7]; Griffin–Ono–Rolen–Zagier [8]; de Bruijn–Newman/Rodgers–Tao [9]; Mostafazadeh [10] and Bender–Brody–Müller [11]; Mathlib [12]. Spectral and adelic perspectives (Berry–Keating, Connes) and the Weil/Li positivity criteria are classical antecedents of the operator and positivity viewpoints used here.

Acknowledgement of limitations (summary)

The verified contribution is a correct, tight, axiom-clean *reduction* of RH to an explicit positivity statement, plus an independent operator reduction, a repaired target-function error, and numerical evidence. RH itself remains open, as stated throughout.

Disclosure of AI-assisted tools

In preparing this work the author used large language model assistants—OpenAI GPT-5.5 and Anthropic Claude Opus 4.8—for drafting and editing of exposition, literature triage, code scaffolding for the numerical scripts, and consistency checking of the manuscript. These tools were used strictly as assistance under the author’s direction. They are not authors and bear no responsibility for the work: all mathematical statements, the Lean 4 formalization and its axiom audit, the high-precision numerical computations, and the final text were specified, checked, and verified by the author, who takes full and sole responsibility for the content, including any errors. No claim of correctness rests on the AI tools; every “verified” result is established by the Lean kernel and the `scripts/safe-verify.sh` discipline as described in the text.

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References

- [1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin Akad., 1859.
- [2] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ -Funktion*, Acta Math. **48** (1926), 305–317.
- [3] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Sel. Math.-Fys. Medd. **7** (1927).
- [4] M. Aissen, I. J. Schoenberg, A. Whitney, *On the generating functions of totally positive sequences I*, J. Anal. Math. **2** (1952), 93–103.
- [5] A. Edrei, *On the generating functions of totally positive sequences II*, J. Anal. Math. **2** (1953), 104–109.
- [6] S. Karlin, *Total Positivity, Vol. I*, Stanford Univ. Press, 1968.
- [7] G. Csordas, T. S. Norfolk, R. S. Varga, *The Riemann Hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541.
- [8] M. Griffin, K. Ono, L. Rolin, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), 11103–11110.
- [9] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6.

- [10] A. Mostafazadeh, *Pseudo-Hermiticity versus PT-symmetry*, J. Math. Phys. **43** (2002), 205–214.
- [11] C. M. Bender, D. C. Brody, M. P. Müller, *Hamiltonian for the zeros of the Riemann zeta function*, Phys. Rev. Lett. **118** (2017), 130201.
- [12] The mathlib Community, *The Lean mathematical library*, Proc. CPP 2020, 367–381.